SCOREBOARD – Hadamard 14348 Paley-I (2026-10-09)

Cutoff: 2026-10-09T16:01:36Z (~12:01PM ET)

Object	Status	max|HHˆT-nI|	Wall (s)	SPK
H_14348 Paley-I (q=14347)	PASS	0	2614.6649	**IN SPK** (Paley-I)
H_14348 Paley-II (q=7173)	FAIL/SKIP	–	–	7173=3ˆ2*797 NOT prime power; ===1 mod 4 but composite
H_14348 Kronecker (4*3587)	FAIL/SKIP	–	–	3587=17*211 ===3 mod 4 odd NOT Hadamard-admissible

Units: 695/695 * row blocks 113/113 * overall_pass=True * wall_total=2654.8907s
Gap 2052: **OPEN** * clay_closed=false * GATE-04 **LOCKED** * conjecture **OPEN**
IN SPK via Paley-I; Miyamoto digression skipped; exclude Miyamoto-only 8948/8644/8516/...; prior SPK 10224+10212+10164+10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884 retained

Next optional: **q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3ˆ271*317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59ˆ2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2ˆ2*59ˆ2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2ˆ2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3ˆ2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2ˆ2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1367 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*2389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2ˆ2*3*2389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13ˆ2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2ˆ2*3*7167 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7057->n=14116 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5*941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2ˆ2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7069->n=14140 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7069 prime ===1 mod 4; Paley-I q=14139=3ˆ2*1571 NOT prime -> FAIL; Kronecker 14140=4*3535; 3535=5*7101 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3535); 14140=2ˆ2*5*7101 ===12 mod**

16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7109->n=14220 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7109 prime ===1 mod 4; Paley-I q=14219=59*241 NOT prime -> FAIL; Kronecker 14220=4*3555; 3555=3^2*79 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3555); 14220=2^2*3^2*79 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7129->n=14260 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7129 prime ===1 mod 4; Paley-I q=14259=3*4753 NOT prime -> FAIL; Kronecker 14260=4*3565; 3565=5*23*31 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3565); 14260=2^2*5*23*31 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | FIRST Paley-II DIRECT after 14348 (and FIRST any direct route after 14348): q2=7177->n=14356 (q2=7177 prime ===1 mod 4; Paley-I q=14355=3^2*5*11*29 NOT prime -> FAIL; Kronecker 14356=4*3589; 3589=37*97 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3589); 14356=2^2*37*97 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII); comes BEFORE the next Paley-I 14388) | next Paley-I DIRECT: q=14387->n=14388 (prime ===3 mod 4; only other prime in 14348..14386 is 14369 ===1 mod 4 -> skipped; Paley-II q2=7193 prime ===1 mod 4 -> ALSO a direct route (deferred note-only; Paley-I is the claim path, as 14244); Kronecker 14388=4*3597; 3597=3*11*109 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3597); 14388=2^2*3*11*109 ===4 mod 16) | no Kronecker-only route in 14352..14384 (note-only, NOT on box): 14352=4*3588, 14368=4*3592, 14384=4*3596 are admissible but H_m has no direct classical Paley route (3587=17*211, 1793=11*163; 3591=3^3*7*19, 1795=5*359; 3595=5*719, 1797=3*599) (or gap 2052 / Track B)